



Semester - III					
Data Structures and Applications Category: Professional Core Course (Common to CS, CD, CY and CI Programs) (Theory & Practical)					
Course Code	:	CS233AI	CIE	:	100+ 50 Marks
Credit distribution L:T:P	:	3:0:1	SEE	:	100+ 50 Marks
Pedagogical Hours (L:T:P:S)	:	45:0:30:45	CIE Duration	:	2+2 Hours
Total credits:	:	4	SEE Duration	:	3+3 Hours
Unit - I					9 Hours
Introduction: Abstract Data Types (ADT), Primitive and Non-primitive Data Types, Linear and Non-linear Data Structures. Static and Dynamic Memory Allocation. Linked Lists: Singly, Doubly, Circular, Operations on Linked Lists: Creation, Insertion, Deletion, Traversal, Searching, Applications of Linked Lists. Self-learning / Extended Topics: Multi-level or hierarchical linked lists, Polynomial representation and arithmetic using linked lists, Sparse matrix representation, Circular linked lists for round-robin scheduling, Managing dynamic collections (train composition, student records, inventory), Merging and splitting linked lists, Reverse a linked list (iterative and recursive approaches)					
Unit - II					9 Hours
Stacks: LIFO concept, abstract data type, Stack operations: push, pop, peek, Implementation using arrays and linked lists. Applications: Expression evaluation and conversion, Parenthesis balancing, Tower of Hanoi Self-learning / Extended Topics: Using stacks to eliminate recursion (iterative problem-solving), Implementing multiple stacks in a single array, Undo/Redo operations in text editors, Backtracking problems (maze solving, N-Queens problem), Implementing custom stack variations (Min/Max Stack, Stack of Plates), Simulating browser history navigation, Stock Span Problem, Reverse a Stack Without Extra Space, Balanced parentheses checking, Largest Rectangle in Histogram					
Unit - III					9 Hours
Queues: FIFO concept, abstract data type, Queue operations: enqueue, dequeue, front, rear. Types: Simple, Circular, Double-Ended, Priority Queues. Implementation using arrays and linked lists. Applications: Scheduling, buffering, simulation problems Self-learning / Extended Topics: Implementing multiple queues in a single array, Circular queue operations and efficiency analysis, Double-ended queues (Deque) and applications, Custom queue applications (printer/job scheduling, CPU task buffering, Josephus problem, round-robin scheduling). Combining stack and queue operations for complex problems, BFS in graphs using queue					
Unit - IV					9 Hours
Trees: N-ary Trees, Binary Trees, Binary Search Trees, Expression Trees, Forests. Properties and conversion of N-ary tree / Forest to Binary tree. Traversals: Preorder, Inorder, Postorder. Binary Search Tree operations: Creation, Insertion, Deletion, Display. Threaded Binary Tree. Self-learning / Extended Topics: Constructing and evaluating expression trees (arithmetic/boolean), Level-order traversal using queues, Finding height/depth, leaf nodes, internal nodes, and node count (recursive and iterative), Lowest Common Ancestor (LCA) problems, Advanced traversal techniques: Morris Traversal (Inorder without recursion/stack), Designing custom trees for real-world hierarchical data (file systems, organizational charts), Expression tree evaluation					
Unit - V					9 Hours



Binary Heap: Definition, Construction, Applications

Advanced Trees: AVL Trees, B Tree, B+ Tree, Splay, Tries

Graphs: Introduction, Properties, Representation (Adjacency Matrix, Adjacency List)

Hashing: Open Hashing, Closed Hashing, Collision and Collision Resolution Strategies. **Self-learning / Extended Topics (Application-Oriented):** Heap applications: Priority Queue, Heap Sort, Top-K elements in datasets, AVL Trees for balanced search, B/B+ Trees for database indexing, Splay Trees for frequently accessed data / cache simulation, Tries for dictionary search, autocomplete, prefix search, Graph applications: Social networks, Pathfinding (BFS/DFS), Cycle detection, Connected components, Hashing applications: Symbol tables in compilers, caching, database indexing, Heap Sort variations, BFS/DFS for shortest path, cycle detection in graphs.

Part A

The following programs can be executed on C/C++/Java or any equivalent tool/language

1. **Practice Programs:** Implementation and execution of following programs to understand basic concept and working of various data structures.

- To solve tower of Hanoi problem.
- To Implement a Stack using an Array
- To Implement a Queue using an Array
- To implement Stack using multiple Queues
- To implement Queue using multiple Stacks
- To Search for an Element in a Linked List
- To reverse a Linked List
- To Detect the Cycle in a Linked List
- To Print Height and Depth of given Binary Tree
- To Implement Binary Search Tree and tree traversals

2. Lab Programs: (At-least two applications from each of the data structures basket)

- Application of List
 - Implementation of sparse matrix multiplication.
 - Implementation of polynomials operations (addition, subtraction) using Linked List.
 - Merge two sorted linked lists.
 - Implementation of Josephus problem using circular linked list.
 - Detect and remove loops in linked lists.
- Application of Stack
 - Implementation of Infix to Postfix conversion
 - Implementation of Infix to Prefix conversion
 - Implementation of evaluation of postfix expression
 - Implementation of evaluation of prefix evaluation
 - Iterative Tower of Hanoi using stack
- Application of Queue
 - Implement Circular Buffer or Ring Buffer
 - Implement Priority Queue to Add and Delete Elements
 - Implementation of Double Ended Queue
 - Implementation of maze problem
 - CPU Round-Robin scheduling simulation
- Application of Trees
 - Implementation of Binary tree and perform level-order traversal using a queue.
 - Implementation of Expression Tree.
 - Implementation of a decision tree for a simple classification problem.
 - Implementation of Binary Search Tree based dictionary
 - Implementation of Threaded Binary Tree (Left-in / Right-in / Double threaded).
- Application of Advanced trees and Hash Table
 - Implementation of Double hashing technique
 - Implementation of AVL tree.
 - Implementation of priority queue using Binary Heap.
 - Implementation of Graph using adjacency matrix and check if it is connected graph.
 - Implementation of dictionary using Trie
 - Implementation of B tree.

Part B

3. **Experiential Learning:** Students are encouraged to implement an advanced data structure not covered in the syllabus, such as a Red-Black Tree, Fibonacci Heap, Segment Tree, Skip List, Suffix Tree, or advanced graph algorithms, and demonstrate one of its applications. They are expected to submit the working code along with a brief report explaining the implementation and its application

Course Outcomes: After completing the course, the students will be able to:

CO1: Demonstrate an understanding of the concepts, characteristics, representations, and operations of fundamental linear and non-linear data structures.

- CO2: Apply appropriate data structures and memory management concepts to organize and manipulate data efficiently.
- CO3: Analyze computational problems and select appropriate data structures and algorithms for efficient solutions.
- CO4: Develop data structure-based solutions for computational problems using good programming practices
- CO5: Develop and demonstrate data structure-based solutions for interdisciplinary and real-world problems, considering societal needs, through experiential learning and self-directed learning.

References

- [1] E. Horowitz et al., *Fundamentals of data structures in C*, 4. [print.] New York: Computer Science Press, 1994, 585 pp., ISBN: 0716782502
- [2] N. Karumanchi, *Data structures and algorithms made easy*, 2nd ed. Charleston, SC: CareerMonk Publications, 2012, 417 pp., Includes bibliographical references, ISBN: 1468108867
- [3] M. A. Weiss, *Data structures and algorithm analysis in C++*, Fourth edition. Boston: Pearson, 2014, 635 pp., ISBN: 9780132847377
- [4] T. H. Cormen, *Introduction to algorithms*, Fourth edition, C. E. Leiserson et al., Eds. Cambridge, Massachusetts: The MIT Press, 2022, 11291 pp., ISBN: 9780262367509
- [5] T. Langsam Augenstein, *Data Structures Using C and C+*. PEARSON INDIA, 2021, ISBN: 9789332549319

RUBRICS FOR THE CONTINUOUS INTERNAL EVALUATION

#	COMPONENTS	MARKS
1	QUIZZES: Quizzes will be conducted in online/offline mode. TWO quizzes will be conducted and each Quiz will be evaluated for 10 Marks. The sum of the two will be the final quiz marks.	20
2	TESTS: Students will be evaluated in test through descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). THREE tests will be conducted (Two regular tests and One optional Improvement test). Each test will be evaluated for 50 Marks, adding upto 100 Marks. Final test marks will be reduced to 40.	40
3	EXPERIENTIAL LEARNING: Students shall be evaluated based on the creativity, innovation, and practical implementation of their proposed solutions. The evaluation shall be conducted in two phases. The implementation modalities for experiential learning activity shall be decided considering the nature of the course and its interdisciplinary requirements. Each phase shall carry 20 marks, with the total evaluation carrying 40 marks.	40
4	LAB: Conduction of laboratory exercises, lab report, observation and analysis (30 Marks), Lab test (10 Marks) and problem based learning review phase I and II (10 Marks) adding upto 50 marks . The final lab marks is for 50.	50
MAXIMUM MARKS FOR THE CIE		150

RUBRICS FOR SEMESTER END EXAMINATION (THEORY)

Q.NO.	CONTENTS	MARKS
PART A		



RUBRICS FOR SEMESTER END EXAMINATION (THEORY) (Continued)		
Q.NO.	CONTENTS	MARKS
1	Objective type questions covering entire syllabus	10
PART B (Maximum of THREE Sub-divisions only)		
2	Unit - I: Compulsory	18
3	Unit - II: Compulsory	18
4 & 5	Unit - III: Question 4 or 5	18
6 & 7	Unit - IV: Question 6 or 7	18
8 & 9	Unit - V: Question 8 or 9	18
TOTAL		100

RUBRICS FOR SEMESTER END EXAMINATION (LAB)		
Q.NO.	CONTENTS	MARKS
1	Evaluation will be conducted as problem / project based learning, which is program / laboratory specific	50
TOTAL		50